

INST327-0303

Citations Management Database

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Team Santiago

Afaan Kamran, Nathan Gomes, Runvir Gabri, Shemar Anglin, Carter Pry

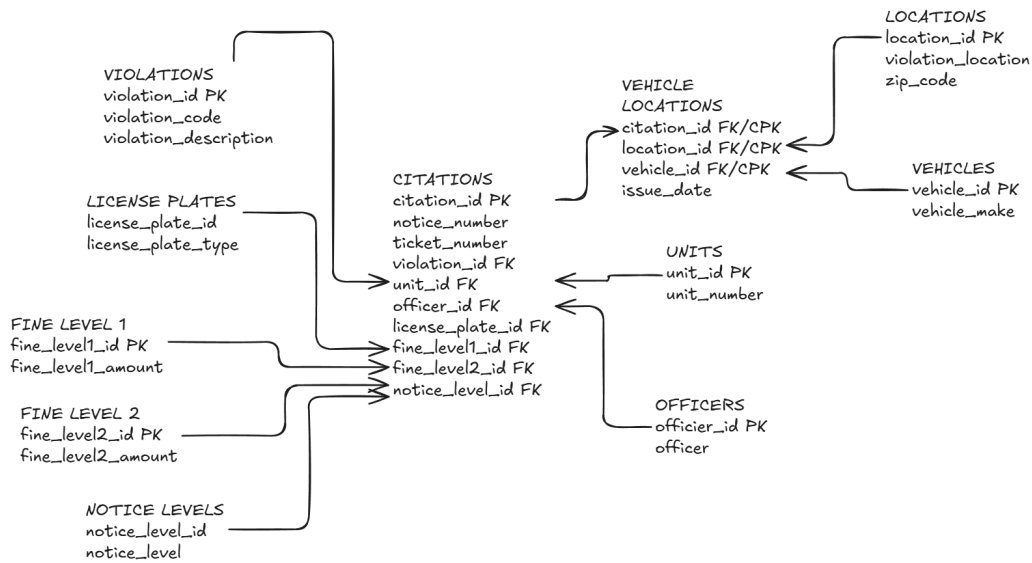
Database Overview

The purpose of our database, the Citations Management Database, is to manage and analyze traffic citations in the city of Chicago. The database consists of data such as different traffic violations, the fine levels of the traffic citations, the notice levels, the different license plates of the citations, the officers, vehicles, and the different locations where traffic citations are originally given or recorded if it was through a camera unit. Potential uses include, but are not limited to improving traffic enforcement strategies, identifying the most common places traffic citations occur, and keeping drivers educated about the different trends and patterns in traffic citations and the opportunities that come with those citations.

Scope Definition

We began by analyzing the large dataset containing Chicago's traffic information. The original dataset contains more information than what is specified for our entities and attributes. However, we considered more essential entities and attributes and disregarded the redundant information. All of this made normalization much easier.

Database Design



The Citations Management database consists of 11 tables:

- Citations: This is the central table linked to all other tables via foreign keys.
- Violations: This table contains the different violation codes and the description of those codes.
- Fine Level 1 & Fine Level 2: These two tables record the two different fine levels that each citation may have.
- Notice Levels: This defines the type of notice issued for the citation.
- License Plates: Contains information about the different license plate types.
- Officers: A record of the officers who issued the citations to drivers.
- Units: The different traffic units that are present for most citations
- Locations: This record includes address and ZIP codes for where the violations occurred.
- Vehicles: A record of the different vehicle makes

Project Evolution

The evolution of our project was straightforward. Initially, our scope and project design were broad and did not include many tables. Our attributes and entities were also not explicit. However, after further analysis of the Chicago Traffic Citations dataset, we saw that more

relationships and tables could be used to further our project goals. Over time, our database structure became more uniform.

Stored Programs

Our stored programs were made to address practical questions that are not able to be interpreted just from analyzing the data alone:

1. Views:

- What are the citation trends by ZIP code?
- What summaries can we gather from the different officers?
- Grouped notice levels for analysis.
- The total fines from the data are based on violation type.

2. Procedures:

- Retrieved citation information based on specific violations and/or vehicle makes.

3. Functions:

- What is the average number of citations per issue date?
- Sums of violations based on vehicle make.

Ethical Considerations

Our database does not have significant ethical considerations. It contains a lot of information about Chicago drivers and is intended only to spread awareness of traffic rules, laws, and citations and to understand trends in these areas. The database is meant to assist Chicago residents in increasing their traffic knowledge and helping reduce the number of citations issued to drivers.

Future Improvements

With unlimited resources, we could hire people with skills for the database to consist of AI and machine learning technologies to provide better utilities for analyzing this data type with much information. This could help users answer more complex questions and better understand the Chicago driving experience.

Team Contributions

The project consisted of a collaborative effort:

- Shemar developed the Excel sheet and designed the 3NF diagram.
- Afaan developed the ERD diagram.
- Nathan dealt with exporting the necessary files
- Runvir compiled the necessary information for the report.